Answers

Worksheet

Securing understanding of place value in integers

Task A: Recognising place value

Below are two separate sets of digit cards.

For each set, use the digit cards to create:

8 7 9 6 5 6 | 0 1 3 5 2 6

1) the largest integer	*987 665*	*653 210*	
2) the smallest integer using all cards	*566 789*	*12 356*	*0 is at the front but is not written*
3) the integer closest to 600	*598*	*601*	
4) the integer closest to 1000	*987*	*1023*	

Another digit card is added to each set at random:

8 7 9 6 5 6 ? | 0 1 3 5 2 6 ?

5) Is there a scenario where the second set of digits could make a larger integer than the first? Why?

No.
E.g. The first set already has the digits 9 and 8 which are larger than all digits in set 2. Even if set 2 received a 9 the largest possible value would only be 9 653 210

6) Jun and Lucas are trying to make the smallest values possible using all of their digits cards.
Identify and correct any mistakes in their statements below.

3 3 5 2 7

Jun

8 0 0 9 8

Lucas

Jun: "The smallest number I can make is 23357"

This should be written with a space. It should be written as 23 357.

Lucas: "The smallest number I can make is 8089"

The number 889 can be made using two leading zeros. (00 889)

Jun: "If I swap my 2 for one of Lucas' 0 cards I can then make the smallest number."

This is false because Lucas will now have the digit 2 so can make 2889 whereas the smallest number Jun can make is 3357

Task B: Integers as sums of their digits' place value

Fill in the blanks:

1) $2372 = 2000 + 300 + 70 + 2$

$= 2 \times 1000 + 3 \times 100 + 7 \times 10 + 2 \times 1$

2) $36\,220 = 30\,000 + 6000 + 200 + 20$

$= 3 \times 10\,000 + 6 \times 1000 + 2 \times 100 + 2 \times 10$

3) $858 = 800 + 50 + 8$

$= 8 \times 100 + 5 \times 10 + 8 \times 1$

4) $5\,003\,009 = 9 + 5\,000\,000 + 3000$

$= 9 \times 1 + 3 \times 1000 + 5 \times 1\,000\,000$

5) $17\,606 = 7000 + 6 + 10\,000 + 600$

$= 7 \times 1000 + 6 \times 1 + 6 \times 100 + 1 \times 10\,000$

6) $300\,280 = 80 + 200 + 300\,000$

$= 8 \times 10 + 2 \times 100 + 3 \times 100\,000$

Task C: Rearranging digits in integers

1) Write down all the unique 3-digit integers you can make with the digits below.

2	1	9

129, 192, 219, 291, 912, 921

2) Choose three digits which can make exactly three unique 3-digit integers.

E.g.

1	2	1

Any three non-zero digits where two digits are the same. E.g. 121

3) Choose three digits which can make exactly one unique 3-digit integer.

E.g.

2	2	2

Any three non-zero digits where all three digits are the same. E.g. 222

4) Write all the unique 4-digit integers which can be made with the digits below.

5	6	6	6

5666, 6566, 6656, 6665

The only unique digit is 5. There are four place value columns the 5 could go in so there are four unique integers.

5) Aisha arranges the digits above to make the largest integer possible. She then takes a zero digit card. Where can she place the zero card so that:

a) her new integer is the largest 5-digit integer possible 66 650

b) her new integer is as close to 61 000 as possible 60 665

c) her new integer is as close to 66 000 as possible 66 065

If we were able to move the other digits around then 66 056 is closer.

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